

Determinants

Definitions

Let $A = [a_{ij}]$ be an $n \times n$ matrix.

- 1 The **sign** of the (i, j) position is $(-1)^{i+j}$.
- 2 Let $A = [a_{ij}]$ be an $n \times n$ matrix. The **ij^{th} minor** of A , denoted as $\text{minor}(A)_{ij}$, is the determinant of the $(n-1) \times (n-1)$ matrix obtained from deleting the i^{th} row and the j^{th} column of A .
- 3 The **(i, j) -cofactor** of an $n \times n$ matrix A is $C_{ij} = (-1)^{i+j} \text{minor}(A)_{ij}$.

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & 1 \\ 1 & -1 & 0 \end{bmatrix}$$

$$\text{Signs : } \begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & 1 \\ 1 & -1 & 0 \end{bmatrix}$$

$$\text{minor}(A)_{21} = \begin{vmatrix} 0 & 2 \\ -1 & 0 \end{vmatrix} = 0 + 2 = 2$$

$$C_{21} = (-1)^{2+1} \text{minor}(A)_{21} = (-1)^3 (2) = -2$$

Cofactor Expansion Along Row i or Column j

If $A = [A_{ij}]$ is an $n \times n$ matrix, then

$$\det A = a_{i1}C_{i1} + a_{i2}C_{i2} + a_{i3}C_{i3} + \cdots + a_{in}C_{in}$$

(cofactor expansion along row i), and

$$\det A = a_{1j}C_{1j} + a_{2j}C_{2j} + a_{3j}C_{3j} + \cdots + a_{nj}C_{nj}$$

(cofactor expansion along column j).

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & 1 \\ 1 & -1 & 0 \end{bmatrix}$$

cofactor expansion along the first row

$$\det A = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13}$$

$$= (1)(-1)^{1+1} \begin{vmatrix} 1 & 1 \\ -1 & 0 \end{vmatrix} + (0)(-1)^{1+2} \begin{vmatrix} 2 & 1 \\ 1 & 0 \end{vmatrix} + (2)(-1)^{1+3} \begin{vmatrix} 2 & 1 \\ 1 & -1 \end{vmatrix}$$

$$= 1 + 0 - 6 = -5$$

cofactor expansion along the third column

$$\det A = a_{13}C_{13} + a_{23}C_{23} + a_{33}C_{33}$$

$$= (2)(-1)^{1+3} \begin{vmatrix} 2 & 1 \\ 1 & -1 \end{vmatrix} + (1)(-1)^{2+3} \begin{vmatrix} 1 & 0 \\ 1 & -1 \end{vmatrix} + (0)(-1)^{3+3} \begin{vmatrix} 1 & 0 \\ 2 & 1 \end{vmatrix}$$

$$= -6 + 1 + 0 = -5$$

Problem

$$\text{Let } A = \begin{bmatrix} 0 & 1 & -2 & 1 \\ 5 & 0 & 0 & 7 \\ 0 & 1 & -1 & 0 \\ 3 & 0 & 0 & 2 \end{bmatrix}. \text{ Find } \det A.$$

Cofactor expansion along 4th row

$$\det A = (3)(-1)^{4+1} \begin{vmatrix} 1 & -2 & 1 \\ 0 & 0 & 7 \\ 1 & -1 & 0 \end{vmatrix} + (2)(-1)^{4+4} \begin{vmatrix} 0 & 1 & -2 \\ 5 & 0 & 0 \\ 0 & 1 & -1 \end{vmatrix}$$

$$= (-3)(7)(-1)^{2+3} \begin{vmatrix} 1 & -2 \\ 1 & -1 \end{vmatrix} + (2)(5)(-1)^{2+1} \begin{vmatrix} 1 & -2 \\ 1 & -1 \end{vmatrix}$$

$$= (21 - 10) \begin{vmatrix} 1 & -2 \\ 1 & -1 \end{vmatrix} = (11)(1) = 11$$

Determinants of a Triangular Matrices

Definitions

- 1 An $n \times n$ matrix A is called **upper triangular** if all entries **below** the main diagonal are zero.
- 2 An $n \times n$ matrix A is called **lower triangular** if all entries **above** the main diagonal are zero.
- 3 An $n \times n$ matrix A is called **triangular** if it is upper triangular or lower triangular.

Theorem

If $A = [a_{ij}]$ is an $n \times n$ triangular matrix, then

$$\det A = a_{11} \times a_{22} \times a_{33} \times \cdots \times a_{nn},$$

i.e., $\det A$ is the product of the entries of the main diagonal of A .

For example for an upper Triangular matrix $A_{3 \times 3}$ we have:

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{22} & a_{23} \\ 0 & 0 & a_{33} \end{bmatrix}$$

Cofactor expansion along the first column

$$\det A = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ 0 & a_{33} \end{vmatrix} = \underline{a_{11} a_{22} a_{33}}$$

How do elementary row operations affect determinants?

Example

Let

$$A = \begin{bmatrix} 2 & 0 & -3 \\ 0 & 4 & 0 \\ 1 & 0 & -2 \end{bmatrix}.$$

Computing $\det A$ by cofactor expansion along row (or column) 2 yields

$$\det A = (4)(-1)^{2+2} \begin{vmatrix} 2 & -3 \\ 1 & -2 \end{vmatrix} = -4$$

Suppose B_1 , B_2 , and B_3 are obtained from A by **interchanging rows 1 and 2**, **multiplying row 3 by -3** , and **adding twice row 1 to row 3**, respectively, i.e.,

$$B_1 = \begin{bmatrix} 2 & 0 & -3 \\ 1 & 0 & -2 \\ 0 & 4 & 0 \end{bmatrix}, B_2 = \begin{bmatrix} 2 & 0 & -3 \\ 0 & 4 & 0 \\ -3 & 0 & 6 \end{bmatrix}, B_3 = \begin{bmatrix} 2 & 0 & -3 \\ 0 & 4 & 0 \\ 5 & 0 & -8 \end{bmatrix}.$$

How do elementary operations affect the determinant?

$$\det B_1 = (4)(-1)^{3+2} \begin{vmatrix} 2 & -3 \\ 1 & -2 \end{vmatrix} = -\det A$$

$$\det B_2 = (4)(-1)^{2+2} \begin{vmatrix} 2 & -3 \\ -3 & 6 \end{vmatrix} = (4)(-1)^{2+2}(-3) \begin{vmatrix} 2 & -3 \\ 1 & -2 \end{vmatrix} = -3 \det A$$

$$\det B_3 = (4)(-1)^{2+2} \begin{vmatrix} 2 & -3 \\ 5 & -8 \end{vmatrix} = (4)(-1)^{2+2}(-16+15) = \det A$$

Theorem

Let A be an $n \times n$ matrix.

- ① If B is obtained from A by exchanging two different rows of A , then $\det B = -\det A$.
- ② If B is obtained from A by multiplying a row of A by a scalar $k \in \mathbb{R}$, then $\det B = k \det A$.
- ③ If B is obtained from A by adding k times one row of A to a different row of A then $\det B = \det A$.
- ④ If two different rows (or columns) of A are identical, then $\det A = 0$. $\longrightarrow$ Follows from ③

An analogous theorem holds for **elementary column operation**. If A is a matrix, then an **elementary column operations** on A is simply the corresponding elementary row operation performed on the transpose of A , A^T .

Row operation on A

$$A = \begin{bmatrix} 2 & 0 & -3 \\ 0 & 4 & 0 \\ 1 & 0 & -2 \end{bmatrix}$$

$\downarrow R_2 \leftrightarrow R_3$

$$B = \begin{bmatrix} 2 & 0 & -3 \\ 1 & 0 & -2 \\ 0 & 4 & 0 \end{bmatrix}$$

$$\det A = (4)(-1)^{2+2} \begin{vmatrix} 2 & -3 \\ 1 & -2 \end{vmatrix} = -4$$

$$\det B = (4)(-1)^{3+2} \begin{vmatrix} 2 & -3 \\ 1 & -2 \end{vmatrix} = 4$$

$$\boxed{\det A = -\det B, \text{ where } A \xrightarrow{R_2 \leftrightarrow R_3} B}$$

Column operation on A^T

$$A^T = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 4 & 0 \\ -3 & 0 & -2 \end{bmatrix}$$

$\downarrow C_2 \leftrightarrow C_3$

$$B^T = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 0 & 4 \\ -3 & -2 & 0 \end{bmatrix}$$

$$\det A^T = (4)(-1)^{2+2} \begin{vmatrix} 2 & 1 \\ -3 & -2 \end{vmatrix} = -4 = \det A$$

$$\det B^T = (4)(-1)^{2+3} \begin{vmatrix} 2 & 1 \\ -3 & -2 \end{vmatrix} = 4 = -\det A^T = -\det A$$

$$\boxed{\det A^T = -\det B^T, \text{ where } A^T \xrightarrow{C_2 \leftrightarrow C_3} B^T}$$

Problem

Suppose A is a 3×3 matrix with $\det A = 2$. What is $\det(-3A)$?

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \quad -3A = \begin{bmatrix} -3a_{11} & -3a_{12} & -3a_{13} \\ -3a_{21} & -3a_{22} & -3a_{23} \\ -3a_{31} & -3a_{32} & -3a_{33} \end{bmatrix}$$

$$\det(-3A) = (-3)^3 \det A = (-27)(2) = -54$$

Theorem

If A is an $n \times n$ matrix and k is any scalar, then

$$\det(kA) = k^n \det A.$$

As an example see the previous problem.

Determinants of Inverses, Products, and Transposes

Theorem

An $n \times n$ matrix A is invertible if and only if $\det A \neq 0$. In this case,
$$\det(A^{-1}) = \frac{1}{\det A}.$$

Statement 1

Statement 2

Theorem

Let A and B be $n \times n$ matrices. Then

$$\det(AB) = \det A \times \det B.$$

Theorem

If A is an $n \times n$ matrix, then the determinant of its transpose is given by

$$\det(A^T) = \det A.$$

Statement ① $\Rightarrow$ Statement ②

If $A_{n \times n}$ is invertible, then we know that $[A|I] \xrightarrow{\text{Elementary operations}} [I|A^{-1}]$. We also know that elementary operations are reversible. Hence A can be obtained from I by performing a sequence of elementary operations. We conclude that $\det A = \alpha \det I = \alpha$, where $\alpha \in \mathbb{R}$ and $\alpha \neq 0$.

Statement ② $\Rightarrow$ Statement ①: We show that if S_1 is NOT true, then S_2 is NOT true.

If A is NOT invertible, then A must have less than n pivot positions. This means the RREF will have at least one zero on its main diagonal and since the RREF is an upper triangular matrix, its determinant is the multiplication of the entries on the main diagonal and hence $\det A = 0$.